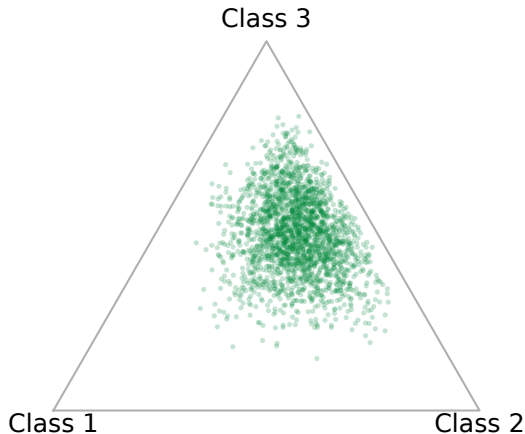


Evidential Bayesian Neural Networks for Uncertainty-Aware Classification

John-Pierre Weideman



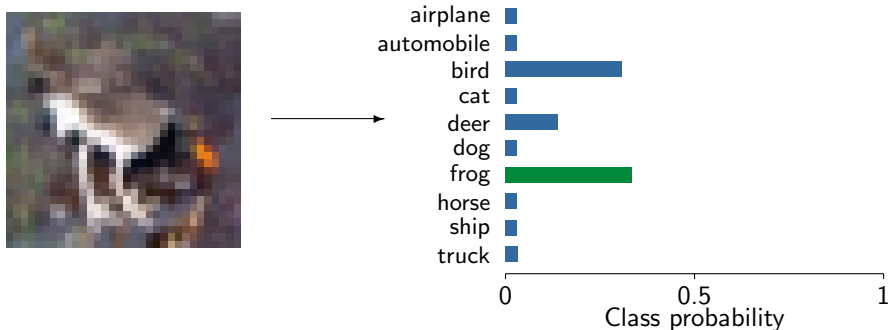
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Why Uncertainty-Aware Modelling Matters

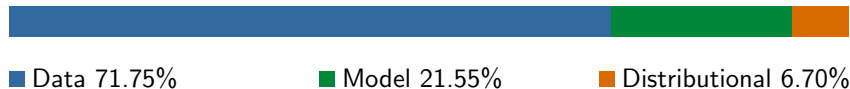
- Neural networks are widely used in vision, language, speech, etc.
- Despite strong performance, their predictions can be overconfident.
- Estimating uncertainty in a model's predictions can support safer decisions.
- Especially important when errors have serious consequences.
 - ▶ Autonomous driving, medical, financial, legal, etc.



- For a classification task, a neural network outputs probabilities for each class label.



- Predictive uncertainty* quantifies uncertainty about the class label of an input.
 - Does not by itself guide decisions about how to act.
 - Where does the predictive uncertainty come from?



- **Aleatoric (data) uncertainty** irreducible uncertainty from complexity in the data.
 - ▶ Defer the decision if high risk.
- **Epistemic (model) uncertainty** reducible uncertainty in the estimated network weights.
 - ▶ Collect more relevant training examples.
- **Distributional uncertainty** is uncertainty about the class probabilities at fixed network weights.
 - ▶ If inputs differ from training data, collect representative training examples.
- **Different uncertainty terms require different actions.**

- **Bayesian neural networks**

- ▶ Model uncertainty over network weights.
- ▶ Aleatoric and epistemic uncertainty.

- **Deterministic Dirichlet networks**

- ▶ Model a distribution over class probabilities
- ▶ Aleatoric and distributional uncertainty.

- Neither decomposes into all three uncertainty terms.

- Model that represents aleatoric, distributional, and epistemic uncertainty.
 - ▶ Combine Bayesian posterior distribution over network weights with a Dirichlet distribution over class probabilities.
- Understand how the resulting predictive uncertainty is divided between the three uncertainty terms.

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- x is an input and $y \in \{1, \dots, C\}$ its label among C classes.
- The network has weights w and outputs the class-probability vector $f(x; w)$.
 - ▶ Cat denotes the categorical distribution over labels.

$$p(y \mid x, w) = \text{Cat}(y \mid f(x; w)) = f_y(x; w).$$

- Training data $D = \{(x_i, y_i)\}_{i=1}^N$ contain N labelled examples.

$$p(D \mid w) = \prod_{i=1}^N p(y_i \mid x_i, w),$$

$$\underbrace{p(w \mid D)}_{\text{posterior}} \propto \underbrace{p(D \mid w)}_{\text{likelihood}} \underbrace{p(w)}_{\text{prior}}.$$

- Predictions average over the posterior over weights.

- A deterministic network uses one fitted set of weights w .
- It outputs concentration parameters $\alpha(x; w)$, with $\alpha_c(x; w) > 0$ for each class c .
- π is a random class-probability vector. Dir denotes its Dirichlet distribution.

$$p(\pi \mid x, w) = \text{Dir}(\pi \mid \alpha(x; w)),$$
$$p(y \mid \pi) = \text{Cat}(y \mid \pi) = \pi_y.$$

- Define the total concentration parameter $\alpha_0(x; w)$ and mean vector $m(x; w)$.

$$\alpha_0(x; w) = \sum_{c=1}^C \alpha_c(x; w), \quad m(x; w) = \frac{\alpha(x; w)}{\alpha_0(x; w)}.$$

- The mean gives the predicted class probabilities, $p(y = c \mid x, w) = m_c(x; w)$.

- Entropy $\mathcal{H}$ measures label uncertainty. $\mathbb{E}$ denotes an average.
- Mutual information $\mathcal{I}$ measures how much knowing another variable reduces label uncertainty, on average.
- $\text{Cat}(y \mid \pi)$ is the categorical label distribution with class probabilities π .

Categorical BNN

$$\underbrace{\mathcal{H}[p(y \mid x, D)]}_{\text{predictive}} = \underbrace{\mathbb{E}_{p(w \mid D)} \left[\mathcal{H}[\text{Cat}(y \mid f(x; w))] \right]}_{\text{aleatoric}} + \underbrace{\mathcal{I}(y; w)}_{\text{epistemic}}.$$

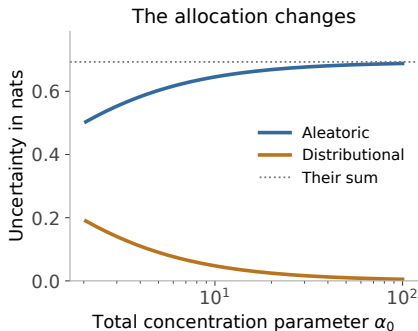
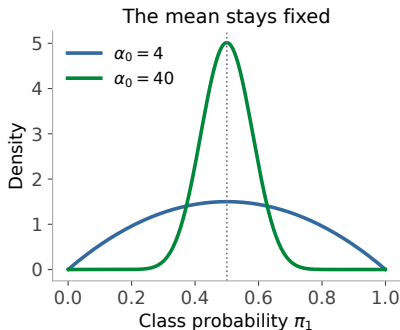
Dirichlet network at fixed weights

$$\underbrace{\mathcal{H}[\text{Cat}(y \mid m(x; w))]}_{\text{predictive}} = \underbrace{\mathbb{E}_{\text{Dir}(\pi \mid \alpha(x; w))} \left[\mathcal{H}[\text{Cat}(y \mid \pi)] \right]}_{\text{aleatoric}} + \underbrace{\mathcal{I}(y; \pi \mid w)}_{\text{distributional}}.$$

Same Prediction, Different Uncertainty Split

- Write α , α_0 , and m for a Dirichlet concentration vector, its total concentration, and its mean.
 - ▶ Multiplying all concentration parameters by the same factor $\lambda > 0$ preserves the mean.

$$\frac{\lambda\alpha}{\lambda\alpha_0} = \frac{\alpha}{\alpha_0} = m.$$



- At the same mean, higher total concentration increases aleatoric uncertainty and decreases distributional uncertainty.

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- The combined model represents the three uncertainty terms.
- Categorical evidential Bayesian neural network (Cat-eBNN)
 - ▶ For input x and data D , the weight posterior $p(w | D)$ gives epistemic uncertainty.

$$p(\pi | x, w) = \text{Dir}(\pi | \alpha(x; w)) \quad \text{distributional uncertainty}$$

$$p(y | \pi) = \text{Cat}(y | \pi) \quad \text{aleatoric uncertainty}$$

- The network gives $\alpha(x; w)$. The random class-probability vector is π .

- At fixed mean vectors, single labels cannot inform the total concentration parameter.

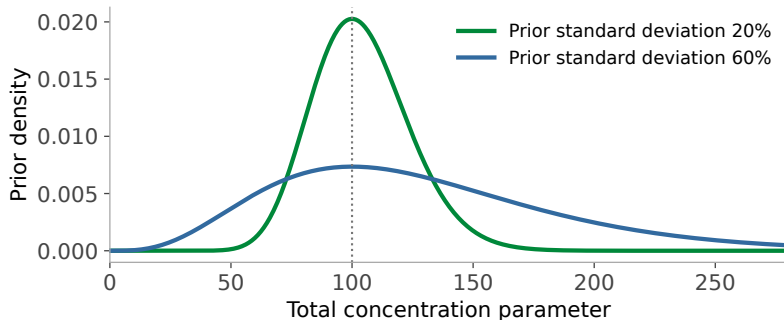
$$p(y \mid x, w) = m_y(x; w), \quad m(x; w) = \frac{\alpha(x; w)}{\alpha_0(x; w)}.$$

$$\frac{\lambda \alpha}{\lambda \alpha_0} = \frac{\alpha}{\alpha_0} = m, \quad \lambda > 0.$$

- Rescaling changes total concentration without changing the mean vector or label likelihood.

An explicit Gamma prior

- The Gamma prior states an assumption about the total concentration parameter.



- Gamma factors act on network outputs at the training inputs.
- Illustrative prior mode 100.
 - ▶ Standard deviations are relative to this mode.

The prior changes the reported allocation

- Higher prior modes correspond to a lower distributional share in these runs.

Gamma prior mode	realised total concentration	aleatoric uncertainty %	distributional uncertainty %
30	30.4 (30.2–30.5)	89.24	8.98
100	99.7 (99.5–99.8)	91.11	4.79
300	291.4 (291.2–291.6)	98.07	0.87

- The trained mean vectors also change across runs.
- CIFAR-10. Means and ranges over three seeds.
 - Final Bayesian model averaging (BMA) over ten samples.

Prediction with single labels

- Negative log likelihood favours the Cat-BNN on both datasets.
- Negative log likelihood (NLL).
 - ▶ Lower is better.
- Accuracy results are mixed.

model	accuracy	NLL
Fashion-MNIST		
Cat-BNN	0.921 (0.917–0.924)	0.221 (0.211–0.230)
Cat-eBNN	0.913 (0.908–0.916)	0.431 (0.410–0.458)
CIFAR-10		
Cat-BNN	0.856 (0.850–0.859)	0.426 (0.418–0.441)
Cat-eBNN	0.867 (0.866–0.868)	0.470 (0.461–0.476)

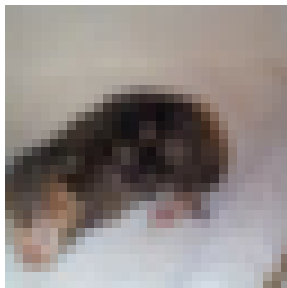
- Means and ranges over three seeds. Final BMA over ten samples.

Outline

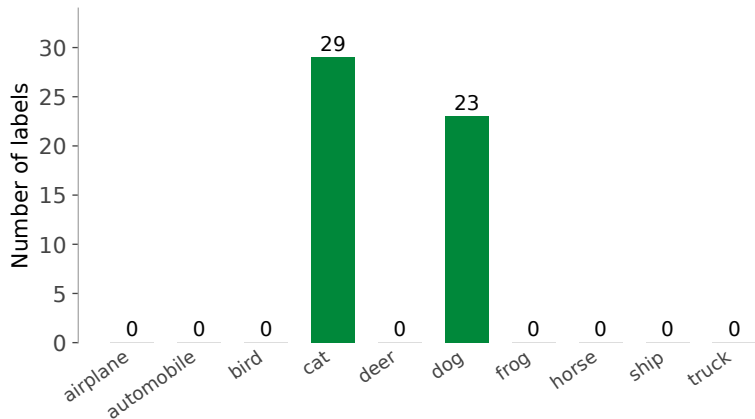
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Repeated labels reveal disagreement

- Retaining repeated labels preserves observed disagreement.



One CIFAR-10H image



- 52 observed human labels for the same input.

A model for complete count vectors

- Repeated labels change the observation model.
- Labels are aggregated by class into a count vector n .
 - ▶ The count total is $R = \sum_c n_c$.

Multinomial Bayesian neural network (M-BNN)

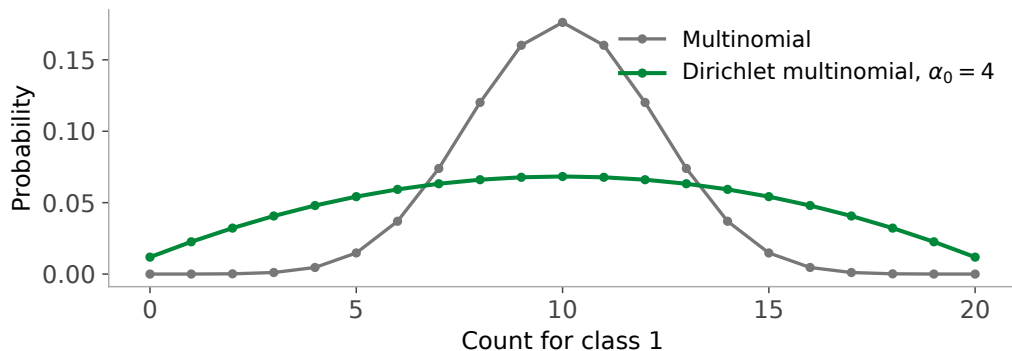
$$p(n \mid x, w) = \text{Mult}(n \mid R, f(x; w)).$$

Dirichlet multinomial evidential Bayesian neural network (DM-eBNN)

$$p(n \mid x, w) = \text{DM}(n \mid R, \alpha(x; w)).$$

- $f(x; w)$ is the class-probability vector of the M-BNN.

- The count likelihood can inform the total concentration parameter.



- The mean vector and count total stay fixed. The count probabilities differ.
 - ▶ Two class illustration with mean vector (0.5, 0.5) and count total 20.

- Additional count variance contributes to better count prediction.
- The mean matched control keeps the DM-eBNN mean vectors and weight samples.
 - ▶ Only the observation model changes to the multinomial distribution.

initialisation	Gamma prior mode	M-BNN count NLL	DM-eBNN count NLL	mean matched control NLL
cold	50	19.161 (18.012–19.754)	12.718 (12.486–13.007)	23.719 (23.195–24.467)
cold	200	19.161 (18.012–19.754)	13.199 (11.872–14.051)	20.291 (17.602–22.035)
warm	50	27.540 (23.630–31.685)	14.392 (13.724–15.529)	28.802 (26.771–32.415)
warm	200	27.540 (23.630–31.685)	16.152 (15.664–16.860)	26.207 (25.281–27.394)

- CIFAR-10H. Means and ranges over three seeds. Lower count NLL is better.

- The prior and initial checkpoint still affect realised total concentration.
- Realised total concentration after sampling on CIFAR-10H.

Gamma prior standard deviation during sampling	pretraining Gamma prior mode 50	pretraining Gamma prior mode 200
20%	51.0 (50.8–51.1)	182.7 (181.5–184.8)
100%	69.8 (69.2–70.3)	162.5 (160.6–165.5)
no Gamma prior	79.3 (78.2–80.1)	155.9 (154.1–159.0)

- Means and ranges over three seeds. Warm initialisation with a fixed sampling budget.

- The DM-eBNN reports the three terms. Their practical value remains open.

model	Gamma prior mode	aleatoric uncertainty %	distributional uncertainty %	epistemic uncertainty %
M-BNN	–	95.25	–	4.75
DM-eBNN	50	87.60	7.71	4.69
DM-eBNN	200	88.28	4.07	7.65

- Shares of each model's own predictive uncertainty in a class label.
 - ▶ CIFAR-10H. Averages over inputs and three seeds. Warm initialisation.

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- The observation model determines what labels can inform.
- Both models represent aleatoric, distributional, and epistemic uncertainty.
- At fixed mean vectors, the Cat-eBNN relies on the prior assumption for total concentration.
- The DM-eBNN count likelihood can also inform total concentration.
- Better count prediction does not validate the uncertainty decomposition.

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- It does not demonstrate improved decisions, and it does not study decision making at all!.
- **One sampling chain per run.**
 - ▶ Test several chains, larger budgets, and other samplers.
- **One dataset with repeated labels.**
 - ▶ Evaluate further datasets with repeated labels.
- **The uncertainty terms need validation.**
 - ▶ Test known label variation and practical decision tasks.

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- Repeated labels can inform the total concentration parameter.
- Both proposed models represent aleatoric, distributional, and epistemic uncertainty.
- The Cat-eBNN states the assumption through a Gamma prior.
- The DM-eBNN improves count prediction in the tested CIFAR-10H settings.
- The practical value of the uncertainty decomposition remains open.

Backup. Decomposition of predictive uncertainty

- x is the fixed input, y the class label, and D the training data.
- w denotes network weights. π is the random class-probability vector.
- $\mathcal{H}$ denotes entropy. $\mathcal{I}$ denotes mutual information.

$$\underbrace{\mathcal{H}[p(y \mid x, D)]}_{\text{predictive}} = \underbrace{\mathbb{E}_{p(w \mid D)} \left[\mathbb{E}_{p(\pi \mid x, w)} [\mathcal{H}[\text{Cat}(y \mid \pi)]] \right]}_{\text{aleatoric}} \\ + \underbrace{\mathbb{E}_{p(w \mid D)} [\mathcal{I}(y; \pi \mid w)]}_{\text{distributional}} + \underbrace{\mathcal{I}(y; w)}_{\text{epistemic}}.$$

- $\text{Cat}(y \mid \pi)$ is the label distribution with class probabilities π .
- Expectations average over $p(w \mid D)$ and $p(\pi \mid x, w)$.
- Mutual information terms also condition on the fixed x and D , omitted for brevity.

- The Gamma factors act on network outputs at the training inputs.

$$\tilde{p}(w) \propto p(w) \prod_{i=1}^N \text{Gamma}(\alpha_0(x_i; w) \mid \kappa, \beta), \quad \kappa > 1, \beta > 0.$$

$$\bar{\alpha}_0 = \frac{\kappa - 1}{\beta} > C.$$

- The Gamma prior standard deviation is $\sqrt{\kappa}/\beta$.
- The Gamma factors use the training inputs but no labels.
- $p(w)$ is the Gaussian weight prior. C counts classes. N counts training inputs.

- Equal mean vectors leave the posterior odds equal to the prior odds.
- Consider weights w and w' with equal mean vectors on every training input.

$$p(D | w) = \prod_{i=1}^N m_{y_i}(x_i; w) = \prod_{i=1}^N m_{y_i}(x_i; w') = p(D | w').$$

$$\frac{p(w | D)}{p(w' | D)} = \frac{\tilde{p}(w)}{\tilde{p}(w')} \frac{p(D | w)}{p(D | w')} = \frac{\tilde{p}(w)}{\tilde{p}(w')}.$$

- The aleatoric and distributional terms have closed forms.
- $\mathcal{H}$ denotes entropy. $\mathcal{I}$ denotes mutual information. ψ is the digamma function.

$$\mathbb{E}_{\text{Dir}(\pi|\alpha(x;w))} [\mathcal{H}[\text{Cat}(y | \pi)]] = \sum_{c=1}^C m_c(x; w) [\psi(\alpha_0(x; w) + 1) - \psi(\alpha_c(x; w) + 1)].$$

$$\mathcal{I}(y; \pi | w) = \mathcal{H}[\text{Cat}(y | m(x; w))] - \mathbb{E}_{\text{Dir}(\pi|\alpha(x;w))} [\mathcal{H}[\text{Cat}(y | \pi)]].$$

- The Cat-eBNN and DM-eBNN average these terms over the weight posterior.

- Finite total concentration permits additional count variance when $R > 1$.
- Both distributions have class-probability vector q and count total R .

$$\mathbb{E}_{\text{Mult}(n|R,q)}[n_c] = \mathbb{E}_{\text{DM}(n|R,\alpha_0 q)}[n_c] = Rq_c.$$

$$\frac{\text{Var}_{\text{DM}(n|R,\alpha_0 q)}[n_c]}{\text{Var}_{\text{Mult}(n|R,q)}[n_c]} = \frac{\alpha_0 + R}{\alpha_0 + 1}.$$

- The variance ratio assumes $0 < q_c < 1$. It is one when $R = 1$.

- Each run uses one chain of stochastic gradient Langevin dynamics (SGLD).

	Single labels	Repeated labels
Final BMA samples	10	40
Cold initialisation epochs	150 / 600	3000
Warm pretraining epochs	50 / 300	400
Warm SGLD epochs	100 / 300	1000

- The paired single label values refer to Fashion-MNIST / CIFAR-10.
- Cold and warm budgets differ in the count experiments.

Single-label predictive performance, test split		
model	accuracy	NLL
Fashion-MNIST		
Softmax	0.928 (0.923–0.933)	0.202 (0.190–0.214)
EDL	0.924 (0.919–0.926)	0.496 (0.477–0.520)
Cat-BNN	0.921 (0.917–0.924)	0.221 (0.211–0.230)
Cat-eBNN	0.913 (0.908–0.916)	0.431 (0.410–0.458)
CIFAR-10		
Softmax	0.879 (0.875–0.882)	0.554 (0.512–0.595)
EDL	0.885 (0.884–0.887)	0.530 (0.517–0.544)
Cat-BNN	0.856 (0.850–0.859)	0.426 (0.418–0.441)
Cat-eBNN	0.867 (0.866–0.868)	0.470 (0.461–0.476)

- Means and ranges over three seeds.
 - ▶ Sampled models use final BMA over ten samples.

- A tighter Gamma prior keeps realised total concentration closer to the prior mode.

Gamma prior standard deviation	realised total concentration	distributional uncertainty %	NLL
1000%	83.7 (82.2–85.0)	6.03	0.473 (0.465–0.477)
60%	97.2 (97.1–97.3)	5.03	0.466 (0.458–0.472)
20%	99.7 (99.5–99.8)	4.79	0.470 (0.461–0.476)
5%	100.1 (100.0–100.1)	4.32	0.508 (0.499–0.516)

- CIFAR-10. Prior mode 100. Means and ranges over three seeds.

model	Gamma prior mode	majority label accuracy	count NLL
Cold initialisation			
M-BNN	–	0.724 (0.712–0.744)	19.161 (18.012–19.754)
DM-eBNN	50	0.712 (0.697–0.731)	12.718 (12.486–13.007)
DM-eBNN	200	0.702 (0.693–0.715)	13.199 (11.872–14.051)
Warm initialisation			
M-BNN	–	0.746 (0.722–0.776)	27.540 (23.630–31.685)
DM-eBNN	50	0.753 (0.720–0.777)	14.392 (13.724–15.529)
DM-eBNN	200	0.773 (0.759–0.781)	16.152 (15.664–16.860)

- CIFAR-10H. Means and ranges over three seeds. Final BMA over 40 samples.

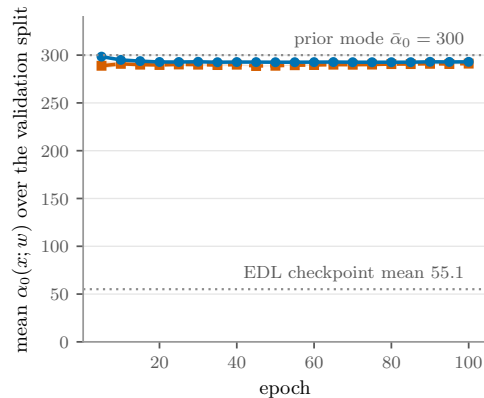
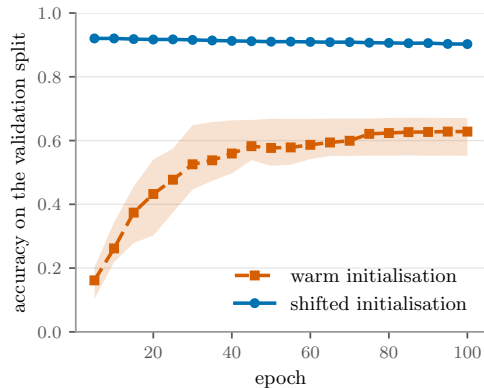
- The controls retain the DM-eBNN mean vectors and weight samples.

initialisation	Gamma prior mode	realised total concentration	mean matched control NLL
cold	50	50.0 (49.6–50.6)	23.719 (23.195–24.467)
cold	200	178.6 (176.4–180.3)	20.291 (17.602–22.035)
warm	50	51.0 (50.8–51.1)	28.802 (26.771–32.415)
warm	200	182.7 (181.5–184.8)	26.207 (25.281–27.394)

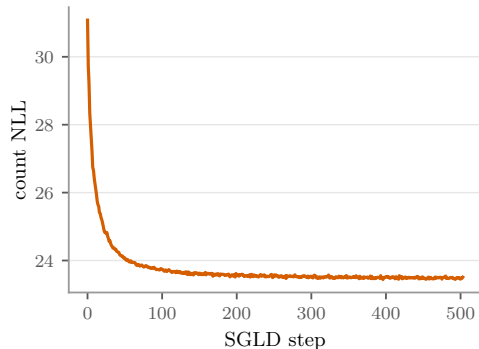
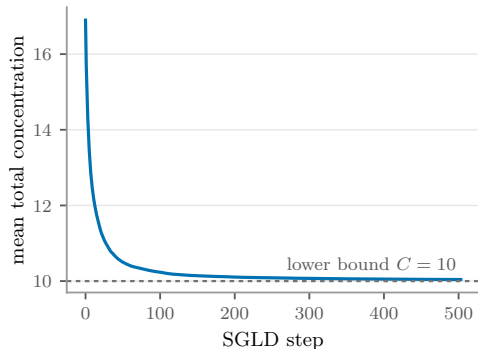
- The control uses the multinomial likelihood.
- CIFAR-10H. Means and ranges over three seeds.

Backup. Initialisation diagnostics

- Initialisation affects predictive performance within the tested budget.

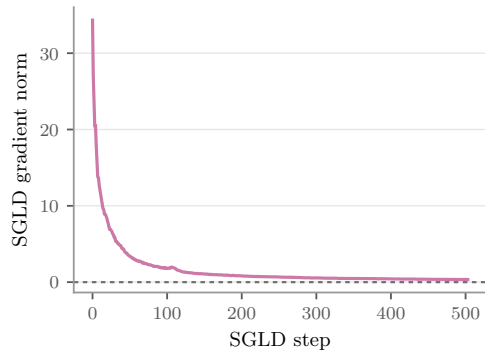
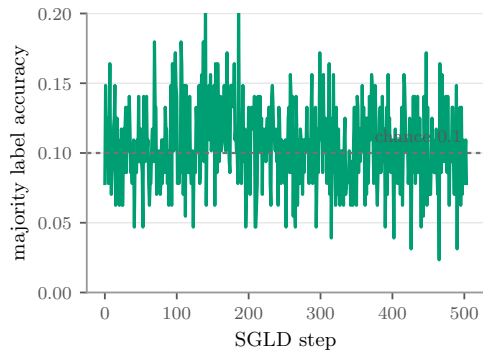


- Fashion-MNIST. Prior mode 300. Current weight checkpoints.
 - ▶ Lines show means over three seeds. Shading shows the seed range.



- One DM-eBNN run on CIFAR-10H. Cold initialisation without the Gamma prior.
 - ▶ Measurements use the current training minibatch before each update.

Backup. Accuracy and gradient norm



- The same DM-eBNN run without the Gamma prior.
 - ▶ The gradient norm falls while majority label accuracy stays near chance.